

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458475.6193 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.0093
Transit depth (Rp/Rs)^2	3.03 +/- 0.32 [%]
Orbital Inclination (inc)	89.27 +/- 1.26
Airmass coefficient 1 (a1)	0.817 +/- 0.017
Airmass coefficient 2 (a2)	0.149 +/- 0.033
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#3 - [275, 208]
Optimal Aperture	3.96
Optimal Annulus	15.74
Transit Duration (day)	0.0775 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.0093 vs 0.1404 (deviation 2.94 σ)
Duration fit vs published	0.0775 d vs 0.0754 d (deviation 0.5 σ)
Mid-time O-C	11.6 min
Depth SNR	15.2
Residual scatter	2.09 %

Light curve

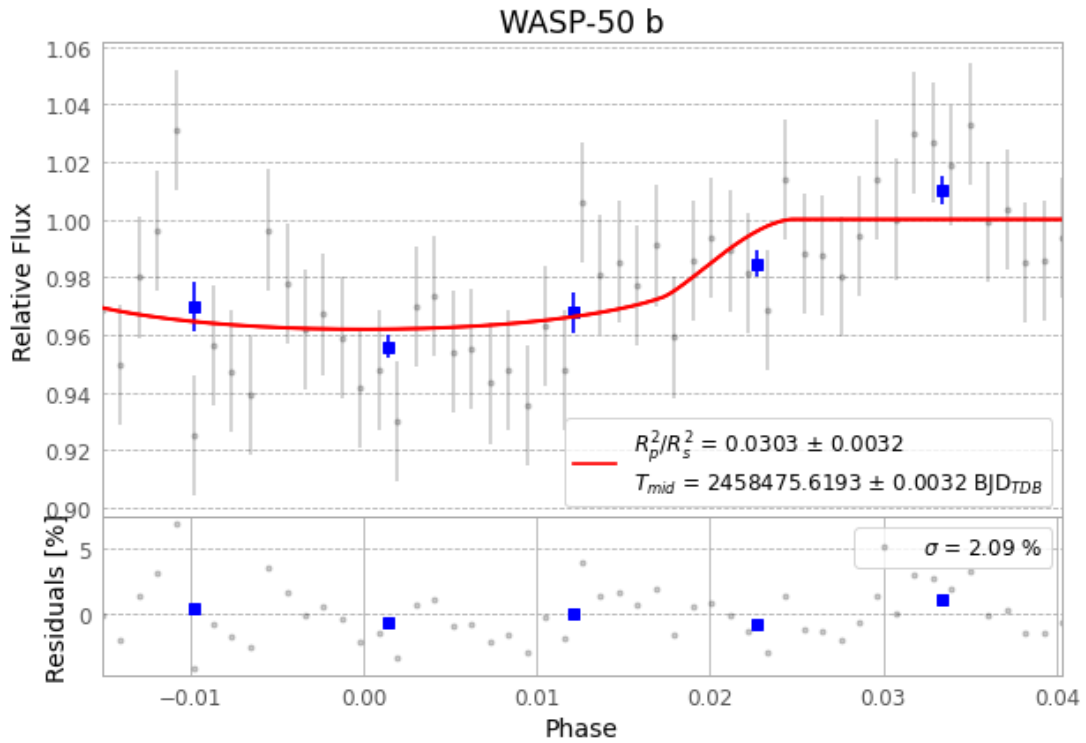


Figure 1 — FinalLightCurve_WASP-50 b_2018-12-22.png (SHA-256 ae88fe2e01d415d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7955efe92ea20cdf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

60 FITS frames (39 MB), WASP-50181223014212.FITS → WASP-50181223043912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-181223.log (68305dc73f9d...), FinalLightCurve_WASP-50 b_2018-12-22.png (ae88fe2e01d4...), FinalLightCurve_WASP-50 b_2018-12-22.csv (b56c42ec5974...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)